

RITHIKA RAMASAMY

FULL STACK & AI ENGINEER | INTELLIGENT SYSTEMS | GENAI & AGENTS

📍 Bengaluru | ☎ +91-8056448057 | [Linkedin](#) | [Github](#) | [Leetcode](#) | [Portfolio](#) | ✉ rithika293.work@gmail.com

SUMMARY

Full-stack developer specializing in React, TypeScript, Node.js, and distributed systems. Delivered scalable fintech workflows and real-time architectures using Redis and Kafka, and shipped production-grade web and mobile applications. Strong in API design, performance optimization, and system reliability, actively building AI-powered applications and intelligent systems.

SKILLS

- Javascript, Python, React, React Native, HTML, CSS3, MongoDB, Redux, MySQL, Supabase, GraphQL, Grpc, Git, REST API, postman, No SQL, Excel, Microservices, GitHub, Node.js, Express.js, Fastify.js, Redis, Kafka, Zod, Visual Studio, Docker, NumPy, Next.JS, TypeScript, Prompt Engineering, Google Colab , YOLOv8, OpenCV, BLE, claude CLI, codex, Multi-system AI Agents, LLMs, RAG, ML, Agents, VectorDB, MCP.

EXPERIENCE

SDE – 1

Neokred Technologies, Bengaluru

Feb 2025 - Present

- Built full-stack features for a high-scale fintech system handling ~10K requests/min using React (TS) and Node.js
- Automated cross-service workflows, reducing manual effort by ~60%
- Implemented Kafka-based producers/consumers for asynchronous and distributed event processing
- Improved performance using Redis caching and optimized API design, reducing latency and enhancing data flow
- Designed idempotent APIs and a configurable logging system, cutting duplicate processing and reducing storage costs by ~34%
- Automated reconciliation workflows, reducing processing time by ~75% and improving system reliability

PROJECTS

Wo-companion (menstrual wellness app prototype)

Apr 2026

🔗 https://github.com/Rithikaram29/women_period_tracker

Python | Scikit-learn | FastAPI | Pandas | React

- Designed and implemented an **AI-driven menstrual wellness system** combining **rule-based cycle phase computation with supervised learning** to estimate fatigue (1–5) and mood from physiological and behavioral inputs.
- Built an **interactive conversational** intake pipeline that captures daily check-ins (sleep, stress, diet, caffeine, mood) and incrementally enriches user state for downstream analysis.
- Engineered a feature transformation layer integrating biological (cycle phase), behavioral (sleep, stress), and lifestyle signals into **structured model-ready representations**.
- Generated a synthetic dataset using **probabilistic modeling and noise injection** to simulate realistic relationships between cycle dynamics, lifestyle factors, and fatigue/mood outcomes.
- Trained and evaluated **Random Forest models (scikit-learn)** to model non-linear feature interactions and produce fatigue and mood predictions.
- Developed a **semantic intent routing layer** using **embedding-based similarity** to classify user queries and dynamically trigger task-specific workflows (e.g., symptom analysis, explanation, lifestyle guidance).
- Implemented a **lightweight agent orchestration layer** that routes intents to modular tools (**prediction models**, nutrition parsing via **RapidFuzz**, and **knowledge retrieval**), aggregates multi-source context (cycle phase, user logs, retrieved data), and composes responses.
- Integrated a **retrieval-augmented generation (RAG)** pipeline with **LLM inference (OpenRouter API)** to generate personalized, context-aware, and safety-constrained explanations.
- Designed a modular, **MCP-compatible tool** interface enabling scalable integration of system components (cycle analysis, user history, RAG retrieval) within an agent-driven architecture.

Go Boarding (Bus ticket booking)

Nov 2024 – Dec 2024

🔗 <https://go-boarding-bus-app-yfk7.vercel.app/>

<https://github.com/Rithikaram29/go-boarding-bus-app>

Typescript | React | MongoDB | Express | Node.js

- Developed a full-stack bus ticket booking platform using the **MERN stack** with **TypeScript**, supporting distinct customer and bus operator workflows.

- Implemented role-based authentication with separate login systems for customers and bus operators using **JWT**, enabling secure and isolated access control.
- Built operator modules allowing bus admins to create, manage, and publish trips dynamically, which are reflected in real-time on the customer booking interface.
- Architected **RESTful APIs with Node.js and Express** for booking workflows, trip scheduling, and seat inventory management.
- Designed optimized **MongoDB schemas** with indexing for efficient querying of trips, bookings, and seat availability.
- Implemented real-time seat availability with concurrency handling to prevent double bookings.
- Developed an interactive seat selection interface in React with synchronized client-server state management.
- Designed responsive UI using **Tailwind CSS**, improving booking experience and reducing user errors.

AI based grocery detection prototype.
Aug 2025

- [🔗 Blog post link](#)
<https://github.com/Rithikaram29/GroceryDetectionApp>
<https://github.com/Rithikaram29/indian-grocery-detection>
- Python | YOLOv8 | MERN | Supabase
- Developed an **AI-powered grocery recognition** system using a **custom-trained YOLOv8 model** to detect multiple grocery items from images.
 - Integrated YOLOv8 directly into the application for end-to-end inference and iterative retraining using evolving datasets.
 - Built a user-in-the-loop annotation tool enabling users to correct misdetections via bounding box labeling in YOLO format (normalized coordinates).
 - Designed a feedback-driven pipeline where user annotations are stored and reused to continuously improve model accuracy.
 - Implemented backend data management using **Supabase (PostgreSQL + object storage)** to store images and structured annotation data.
 - Developed a **full-stack interface (MERN)** for image upload, prediction visualization, and annotation workflows.
 - Engineered a continuous learning loop: inference → user correction → data storage → retraining.
 - Leveraged AI-assisted development tools (ChatGPT, Claude, Bolt.ai) to accelerate frontend development and system integration.
 - Initiated migration to a **React Native mobile app** to enable real-world data collection through on-device image capture.

Eleven11 App (live)
Nov 2025 – Mar 2026

- [🔗 https://play.google.com/store/apps/details?id=com.eleven11studios.eleven11&pcampaignid=web_share](https://play.google.com/store/apps/details?id=com.eleven11studios.eleven11&pcampaignid=web_share)
- Flutter | BLE | Supabase
- Developed application logic and **BLE communication workflows** to interface with offline **IoT hardware** devices, enabling direct device-to-app interaction without internet dependency.
 - Implemented Bluetooth Low Energy (BLE) functionalities including device discovery, connection lifecycle management, and reliable data exchange with embedded hardware.
 - Designed relational schemas in **Supabase (PostgreSQL)** to manage device ownership and user-device mappings.
 - Implemented **application-level validation** to ensure unique device-user association by filtering out already-registered devices during BLE scan and onboarding flows.
 - Designed data synchronization flows** to persist device-generated data to Supabase when connectivity is available.
 - Leveraged **Supabase real-time features** to enable live updates and state synchronization across clients.
 - Contributed to frontend development by integrating BLE-driven data with UI components and handling application state.
 - Structured end-to-end data flow between **hardware, application layer, and backend services** for reliable system behavior.
 - Collaborated with hardware engineers and stakeholders to align integration requirements and ensure smooth project execution.

EDUCATION

Data science and AI, IIT madras Pravathak	Present
Full stack development, Masai Schools	2024
B.Arch, Measi academy of architecture	2022

LANGUAGES

English	Advanced
Tamil	Native